

5.1 Eigenvalues and Eigenvectors

If A is $n \times n$ matrix, then a non-zero vector $\mathbf{x}$ in R^n is called an eigenvector of A if $A\mathbf{x}$ is a scalar multiple of $\mathbf{x}$ that is

$$A\mathbf{x} = \lambda\mathbf{x}$$

$$\Rightarrow A\mathbf{x} - \lambda\mathbf{x} = 0$$

$$\Rightarrow (A - \lambda I)\mathbf{x} = 0$$

For some scalar λ . The scalar λ is called an eigenvalue of A . $\mathbf{x}$ is said to be an eigenvector corresponding to λ .

Example1: Eigenvector of a 2×2 Matrix

Let us check that vector $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is an eigenvector of

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

corresponding to the eigenvalue $\lambda = 3$.

Since

$$A\mathbf{x} = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3\mathbf{x}.$$

Hence the result.

THEOREM 5.1.1 If A is $n \times n$ matrix, then λ is an eigenvalue of A if and only if it satisfies the equation

$$\det(A - \lambda I) = 0 \quad (1)$$

This is called the **characteristic equation** of A .

Note: In Example 1 we observed that $\lambda = 3$ is an eigenvalue of the matrix

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

but we did not explain how we found it. Use the characteristic equation to find all eigenvalues of this matrix.

Example 2: Find eigenvalues of the matrix A

$$\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

Solution: The eigenvalues of A are the solution of the equation $\det(A - \lambda I) = 0$.

As

$$A - \lambda I = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3 - \lambda & 0 \\ 8 & -1 - \lambda \end{bmatrix}$$

$$\Rightarrow \det(A - \lambda I) = \begin{vmatrix} 3 - \lambda & 0 \\ 8 & -1 - \lambda \end{vmatrix}$$

Therefore, the characteristic equation $\det(A - \lambda I) = 0$ is

$$\begin{vmatrix} 3 - \lambda & 0 \\ 8 & -1 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (3 - \lambda)(-1 - \lambda) - 8(0) = 0$$

$$\Rightarrow -(3 - \lambda)(1 + \lambda) = 0$$

$$\lambda = 3, \lambda = -1$$

So, eigenvalues are $\lambda = 3$ & $\lambda = -1$.

Example 3: Find eigenvalues of the matrix A

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix}$$

Solution: The characteristic polynomial of A is

$$\det(A - \lambda I) = \det \left(\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)$$

$$= \det \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 4 & -17 & 8 - \lambda \end{bmatrix} = \begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 4 & -17 & 8 - \lambda \end{vmatrix}$$

$$\begin{aligned}
&= -\lambda \begin{vmatrix} -\lambda & 1 \\ -17 & 8-\lambda \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 4 & 8-\lambda \end{vmatrix} + 0 \begin{vmatrix} 0 & -\lambda \\ 4 & -17 \end{vmatrix} \\
&= -\lambda[-\lambda(8-\lambda) + 17] - 1[0(8-\lambda) - 4(1)] + 0 \\
&= -\lambda[8\lambda + \lambda^2 + 17] - 1[-4] \\
&= -\lambda^3 + 8\lambda^2 - 17\lambda + 4
\end{aligned}$$

To find eigenvalue put $\det(A - \lambda I) = 0$,

$$-\lambda^3 + 8\lambda^2 - 17\lambda + 4 = 0 \quad \Rightarrow \quad \lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0$$

$\lambda = 4$ is the one solution so by synthetic division,

$$\begin{array}{r|rrrr}
4 & 1 & -8 & 17 & -4 \\
& & 4 & -16 & 4 \\
\hline
& 1 & -4 & 1 & 0
\end{array}$$

$$\lambda^2 - 4\lambda + 1 = 0$$

from quadratic equation,

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)}$$

$$\lambda = \frac{4 \pm \sqrt{16 - 4}}{2} \Rightarrow \lambda = \frac{4 \pm \sqrt{12}}{2}$$

$$\lambda = 2 \pm \sqrt{3}$$

So the eigenvalues of A are

$$\lambda = 4, \quad \lambda = 2 + \sqrt{3}, \quad \lambda = 2 - \sqrt{3}$$

Exercise: Find the eigenvalues of the following matrices.

a) $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$

$$\text{b) } \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} -2 & -7 \\ 1 & 2 \end{bmatrix}$$

$$\text{d) } \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{e) } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{f) } \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$\text{g) } \begin{bmatrix} 3 & 0 & -5 \\ 1/5 & -1 & 0 \\ 1 & 1 & -2 \end{bmatrix}$$

$$\text{h) } \begin{bmatrix} -2 & 0 & 1 \\ -6 & -2 & 0 \\ 19 & 5 & -4 \end{bmatrix}$$

$$\text{i) } \begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}$$

Example 4: Find the eigenvalues and the corresponding eigenvectors of the following matrix.

$$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$$

Solution: The eigenvalue/s of A are the solution of the equation

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 3 - \lambda & 0 \\ 8 & -1 - \lambda \end{vmatrix}$$

$$-(3 - \lambda)(1 + \lambda) = 0$$

$$\lambda = 3, \quad \lambda = -1$$

If for $\lambda = 3$, eigenvector is $\begin{bmatrix} x \\ y \end{bmatrix}$, then

$$A\vec{x} = \lambda\vec{x}$$

$$(A - \lambda I)\vec{x} = \vec{0}$$

Using $\lambda = 3$ in

$$\begin{bmatrix} 3 - \lambda & 0 \\ 8 & -1 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 8 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$0x + 0y = 0$$

$$8x - 4y = 0$$

$$8x = 4y \quad \Rightarrow \quad x = \frac{1}{2}y$$

Let $y = t$,

$$x = \frac{1}{2}t$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t \\ t \end{bmatrix} = t \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$

So, $\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = 3$.

Now using $\lambda = -1$,

$$\begin{bmatrix} 4 & 0 \\ 8 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$4x + 0y = 0 \quad \dots (1)$$

$$8x - 0y = 0 \quad \dots (2)$$

$$\Rightarrow \quad x = 0 \text{ from both equations (1) and (2)}$$

As y is free variable so put $y = t$, and $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

So, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = -1$.

Example 5: Find the eigenvalues and the corresponding eigenvectors of the matrix

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

Solution: The eigenvalues of A are the solution of the equation

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 2 & 1 \\ 1 & 3 - \lambda & 1 \\ 1 & 2 & 2 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda) \begin{vmatrix} 3 - \lambda & 1 \\ 2 & 2 - \lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 1 \\ 1 & 2 - \lambda \end{vmatrix} + 1 \begin{vmatrix} 1 & 3 - \lambda \\ 1 & 2 \end{vmatrix} = 0$$

$$(2 - \lambda)[(3 - \lambda)(2 - \lambda) - 2] - 2[2 - \lambda - 1] + 1(2 - 3 + \lambda) = 0$$

$$(2 - \lambda)(6 - 3\lambda - 2\lambda + \lambda^2 - \lambda) - 2[1 - \lambda] + (-1 + \lambda) = 0$$

$$(2 - \lambda)(\lambda^2 - 5\lambda + 4) - 2(1 - \lambda) + (\lambda - 1) = 0$$

$$2\lambda^2 - 10\lambda + 8 - \lambda^3 + 5\lambda^2 - 4\lambda - 2 + 2\lambda + \lambda - 1 = 0$$

$$-\lambda^3 + 7\lambda^2 - 11\lambda + 5 = 0$$

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

$\lambda = 1$ is one solution of above equation because

$$1 - 7 + 11 - 5 = 0$$

$$0 = 0$$

So by using synthetic division we find

$$\begin{array}{r|rrrr} 1 & 1 & -7 & 11 & -5 \\ & & 1 & -6 & 5 \\ \hline & 1 & -6 & 5 & 0 \end{array}$$

$$\lambda^2 - 6\lambda + 5 = 0$$

$$\lambda^2 - 5\lambda - \lambda + 5 = 0$$

$$\lambda(\lambda - 1) - 5(\lambda - 1) = 0$$

$$(\lambda - 1)(\lambda - 5) = 0$$

$$\lambda = 1, \lambda = 5$$

So eigenvalues of A are

$$\lambda = 1, \lambda = 1, \lambda = 5$$

For $\lambda = 1$, eigenvector is

$$(A - \lambda I)\vec{x} = \vec{0}$$

$$\begin{bmatrix} 2 - \lambda & 2 & 1 \\ 1 & 3 - \lambda & 1 \\ 1 & 2 & 2 - \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Put $\lambda = 1$,

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 2y + z = 0$$

$$x + 2y + z = 0$$

$$x + 2y + z = 0$$

Actually, there is only one equation

$$x + 2y + z = 0$$

$$\text{So } x = -2y - z$$

Since y & z are free variables, So put $y = s, z = t$

$$\Rightarrow x = -2s - t$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2s - t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

So $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ are eigenvectors for $\lambda = 1$.

Find the eigenvectors for $\lambda = 5$.

Exercise: Find the eigenvectors for the eigenvalues of the following matrices.

a) $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$

b) $\begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

c) $\begin{bmatrix} 3 & 0 & -5 \\ 1/5 & -1 & 0 \\ 1 & 1 & -2 \end{bmatrix}$

d) $\begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}$

Exercise Set 5.1

In Exercises 1–2, confirm by multiplication that $\mathbf{x}$ is an eigenvector of A , and find the corresponding eigenvalue.

1. $A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix}; \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

Answer:

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2. $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}; \mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

3. Find the characteristic equations of the following matrices:

(a) $\begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$

(b) $\begin{bmatrix} 10 & -9 \\ 4 & -2 \end{bmatrix}$

THEOREM 5.1.2

If A is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), then the eigenvalues of A are the entries on the main diagonal of A .

EXAMPLE 5 Eigenvalues of a Lower Triangular Matrix

By inspection, the eigenvalues of the lower triangular matrix

$$A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ -1 & \frac{2}{3} & 0 \\ 5 & -8 & -\frac{1}{4} \end{bmatrix}$$

are $\lambda = \frac{1}{2}$, $\lambda = \frac{2}{3}$, and $\lambda = -\frac{1}{4}$.